Amrita Vishwa Vidyapeetham

Amrita School of Engineering, Amritapuri Campus

Department of Computer Science and Engineering

Course Plan (June - Nov 2026)

S7 BTech. CSE

1. **Course Information**

Course Code/Title: 23CSE401 Fundamentals of Artificial Intelligence

L-T-P-C: 2-0-2-3

Academic Year and Term: 2026-2027 Odd Semester

Program/Batch/Semester: B. Tech CSE / Semester VII

2. **Course Mentor**: Dr. Namitha K

3. **Course Objectives**

- To introduce classical AI, problem formulation and intelligent agents.
- To introduce techniques for problem solving by search, knowledge-representation and reasoning.
- To introduce game playing and algorithms associated with it.
- To introduce planning, acting, and multi-agent systems

4. **Course Outcomes (CO)**

CO#	Outcome
CO1	Understand the foundations of AI systems, intelligent agents, problem solving strategies and Ethics in AI.
CO2	Apply elementary principles of AI like search and game playing for problem solving.
CO3	Analyse real world problems under uncertainty and solve them using knowledge representation and reasoning.
CO4	Apply planning and decision making in intelligent systems.

5. **CO-PO Affinity Map**

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	–	–	–	–	–	–	2	–	–	–	–	3	2
CO2	3	2	2	–	2	–	–	2	2	2	–	–	3	2
CO3	3	2	2	2	2	–	–	2	2	2	–	–	3	2
CO4	3	2	2	–	1	–	–	2	2	2	–	–	3	2

3-strong, 2-moderate, 1-weak

6. **CO-PO Affinity Justification**

CO	PO/PSO	Affinity	Justification
CO1	PO1	2	CO1 develops foundational knowledge of Artificial Intelligence, intelligent agents, problem formulation, and search strategies, thereby strengthening engineering and computing fundamentals.
CO1	PO8	2	CO1 introduces Ethics in AI, enabling students to understand professional and ethical responsibilities associated with intelligent systems.
CO1	PSO1	3	CO1 strongly supports the application of computing fundamentals for designing intelligent systems and solving computational problems.
CO1	PSO2	2	CO1 provides a foundation for developing AI-based applications in emerging computing domains.
CO2	PO1	3	CO2 applies AI search algorithms and game-playing techniques, strengthening knowledge of computing and engineering principles.
CO2	PO2	2	CO2 enables students to analyze problems and identify appropriate AI search techniques for effective solutions.
CO2	PO3	2	CO2 supports the design of AI-based solutions using heuristic search, game playing, and optimization strategies.
CO2	PO5	2	CO2 promotes the use of modern AI techniques and computational tools for solving complex problems.
CO2	PO8	2	CO2 encourages responsible application of AI algorithms in practical problem-solving scenarios.
CO2	PO9	2	CO2 introduces cooperative learning and swarm intelligence concepts that emphasize collaborative problem solving.

CO2	PO10	2	CO2 develops logical reasoning and communication of AI-based solutions through case studies and applications.
CO2	PSO1	3	CO2 strongly supports the development of intelligent algorithms and computational problem-solving skills.
CO2	PSO2	2	CO2 enables implementation of AI search and game-playing techniques in real-world applications.
CO3	PO1	3	CO3 strengthens understanding of knowledge representation, logical reasoning, and uncertainty handling in intelligent systems.
CO3	PO2	2	CO3 develops analytical skills for solving real-world problems under uncertainty using AI techniques.
CO3	PO3	2	CO3 supports the design of knowledge-based systems and intelligent decision-support applications.
CO3	PO4	2	CO3 encourages investigation and evaluation of reasoning mechanisms such as Bayesian Networks, Fuzzy Logic, and Predicate Logic.
CO3	PO5	2	CO3 promotes the use of modern AI tools and techniques for knowledge representation and reasoning.
CO3	PO8	2	CO3 encourages responsible use of AI reasoning systems in societal and decision-making contexts.
CO3	PO9	2	CO3 involves collaborative analysis of intelligent systems through case studies and practical applications.
CO3	PO10	2	CO3 develops the ability to communicate AI-based reasoning approaches and decision-making strategies effectively.
CO3	PSO1	3	CO3 strongly supports the development of intelligent computing solutions using reasoning and knowledge representation techniques.
CO3	PSO2	2	CO3 enables the implementation of AI-based decision-support and recommendation systems.
CO4	PO1	3	CO4 develops understanding of planning algorithms and intelligent decision-making techniques in AI systems.

CO4	PO2	2	CO4 enables analysis of planning problems and selection of suitable planning strategies for intelligent agents.
CO4	PO3	2	CO4 supports the design of intelligent planning and multi-agent systems for real-world applications.
CO4	PO5	1	CO4 introduces planning frameworks and AI tools used for intelligent decision-making.
CO4	PO8	2	CO4 promotes responsible development and deployment of autonomous planning systems.
CO4	PO9	2	CO4 incorporates multi-agent planning concepts that require teamwork and coordination among intelligent agents.
CO4	PO10	2	CO4 develops communication and documentation skills through analysis of planning applications and case studies.
CO4	PSO1	3	CO4 strongly contributes to designing intelligent systems capable of planning and autonomous decision making.
CO4	PSO2	2	CO4 enables development of AI-driven applications involving planning, reasoning, and intelligent control.

7. **Course Syllabus**

Unit 1

History and Foundations of AI, Introduction to AI and systems - search strategies, problem characteristics, system characteristics, Intelligent Agents – Agents and environments, nature of environments and structure of agents, Ethics in AI; Case study: Games and puzzles.

Unit 2

Problem Solving by Search: Informed search - Heuristic, Hill climbing, Best first search, A*, AO*, Approaches in knowledge representation, Game Playing - minmax algorithm, Alpha beta pruning, Swarm intelligence, cooperative learning – constrained and unconstrained problems; Case study - Example: Group of Drones.

Unit 3

Knowledge representation - Game playing – Knowledge representation, Knowledge representation using Predicate logic, Introduction to predicate calculus, Resolution, Use of predicate calculus, Knowledge representation using other logic-Structured representation of knowledge. Knowledge inference - Production based system, Frame based system. Inference Backward chaining, Forward chaining, Rule value approach, Fuzzy reasoning – Certainty factors, Bayesian Theory-Bayesian NetworkDempster – Shafer theory Case Study-Examples: Recommendation systems, Knowledge based systems in application areas of computer science, Decision support systems, prediction and warning systems.

Unit 4

Classical Planning: Algorithms for Planning, Planning Graphs, Hierarchical Planning, Planning and Acting in Nondeterministic Domain, Map building, Multi-Agent Planning; Case study - Examples: vacuum cleaner, washing machine.

8. **Textbook/References**

Textbook(s) Russell, Stuart, Jonathan, Norvig, Peter, Davis, Ernest. "Artificial Intelligence: A Modern Approach". United Kingdom: Pearson, 4 th Edition, 2022 Deepak Khemani. "A First Course in Artificial Intelligence". McGraw Hill Education (India), 2017

Reference(s) Denis Rothman. "Artificial Intelligence by Example", Packt, 2nd Edition, 2018. Rich and Knight, "Artificial Intelligence",3rd Edition, Tata Mc Graw Hill, 2017.

9. **Evaluation Policy 70:30**

Sl. No.	Assessment	Weightage (%)
1	**Mid Term Examination (50 Marks)**	20
2	**CA – Theory**	10
	Quiz 1 (5 Marks), Quiz 2 (5 Marks)	
3	**CA – Lab**	40
	Lab Submission (5 Marks), Lab Viva (10 Marks)	

	Case Study (25 Marks)	
	Abstract (3 Marks), Presentation (6 Marks),Implementation (6 Marks),Viva (5 Marks), Report (5 Marks)	
4	**End Semester Examination (50 Marks)**	30
	Total	**100**

10. Direct Assessment Tools

. No	Direct Assessment Tools	Weightage (%)	Max Exam Marks	CO1	CO2	CO3	CO4
1	Midterm Examination	20	50	✓	✓		
2	Quiz 1	5	15	✓			
3	Quiz 2	5	15		✓		
4	Lab Submission	5	5	✓	✓	✓	✓
5	Lab Viva	10	10	✓	✓	✓	✓
6	Case Study	25	25			✓	✓
7	End Semester Examination	30	50	✓	✓	✓	

11. CO Attainment Levels

CO	Threshold %	Target %
CO1	60	70
CO2	60	70
CO3	60	70
CO4	60	70

12. Course Delivery Plan

Week	Lect. Hrs.	Topics	Subtopics	CO
1	1	Introduction to AI	History and Foundations of AI	CO1
	2	AI Systems	Introduction to AI Systems, Search Strategies	CO1
2	3	Problem Formulation	Problem Characteristics	CO1
	4	Problem Formulation	System Characteristics	CO1
		Case Study Topic Selection		
3	5	Intelligent Agents	Agents and Environments	CO1
	6	Intelligent Agents	Nature of Environments, Structure of Agents	CO1
4	7	Intelligent Agents	Rational Agents and Agent Architectures	CO1
	8	Ethics in AI	Responsible AI, Ethical Issues in AI	CO1
5	9	Informed Search	Heuristics and Evaluation Functions	CO2
	10	Search Techniques	Hill Climbing Algorithm	CO2
6	11	Search Techniques	Best First Search	CO2
	12	Search Techniques	A* Search Algorithm	CO2
7	13	Search Techniques	AO* Search Algorithm	CO2
	14	Game Playing	Minimax Algorithm	CO2
8	15	Game Playing	Alpha-Beta Pruning	CO2
	16	Swarm Intelligence	Cooperative Learning, Constrained Problems	CO2
		Quiz 1		
		Midterm Examination August 2026		

9	17	Swarm Intelligence	Unconstrained Problems, Group of Drones Case Study	CO2
	18	Knowledge Representation	Introduction and Need for Knowledge Representation	CO3
10	19	Predicate Logic	Predicate Logic and Predicate Calculus	CO3
	20	Predicate Logic	Resolution Principle and Applications	CO3
11	21	Knowledge Representation	Structured Representation of Knowledge	CO3
	22	Knowledge Inference	Production-Based Systems	CO3
12	23	Knowledge Inference	Frame-Based Systems	CO3
	24	Inference Mechanisms	Forward Chaining and Backward Chaining	CO3
13	25	Uncertainty Reasoning	Rule Value Approach and Certainty Factors	CO3
	26	Uncertainty Reasoning	Fuzzy Reasoning	CO3
14	27	Bayesian Reasoning	Bayesian Theory and Bayesian Networks	CO3
	28	Bayesian Reasoning	Dempster-Shafer Theory	CO3
15	29	AI Applications	Recommendation Systems, Knowledge-Based Systems, Decision Support Systems	CO3
	30	AI Applications	Prediction and Warning Systems	CO3
16	31	Planning	Classical Planning, Planning Algorithms	CO4
	32	Planning	Planning Graphs, Hierarchical Planning	CO4
17	33	Intelligent Planning	Planning and Acting in Nondeterministic Domains	CO4

	34	Multi-Agent Systems	Map Building, Multi-Agent Planning	CO4
		Quiz 2		
18	35	**Case Studies Presentations**		CO4
	36	Revision	Unit IV Revision and Case Study Presentations	CO4
End Semester examinations				

13. Faculty Information

Sl. No.	Faculty Name	Class	Signature
1	**Dr. Anjali T**	CSE A	
2	**Dr. Anjali T**	CSE B	
3	**Dr. Namitha K**	CSE C	
4	**Dr. Namitha K**	CSE D	

Rubrics for Case Study

Criteria	Good	Average	Poor
Abstract	Clearly explains the problem, idea, and proposed solution.	Problem and solution are explained but some details are missing.	Problem and solution are not clearly explained.
Presentation	Presentation is clear, organized, and easy to understand.	Presentation is understandable but lacks clarity in some parts.	Presentation is not organized and difficult to understand.
Implementation	Most of the required features are implemented and working correctly.	Some features are implemented, but improvements are needed.	Implementation is incomplete or not working properly.
Viva	Able to explain the work and answer questions confidently.	Able to answer some questions but lacks clarity in explanations.	Unable to explain the work or answer questions satisfactorily.
Report	Report is well organized and covers all important aspects of the work.	Report covers the work but lacks details in some sections.	Report is incomplete or poorly organized.